

Algebra 1
Unit 3
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

A line passes through the points $(-3, 7)$ and $(4, -2)$. Use this information for questions 1 through 3.

(x_1, y_1) (x_2, y_2)

1. Find the slope of the line. Show your work.

$$\frac{-2 - 7}{4 - (-3)} = \frac{-9}{7}$$

2. Write the equation of the line in point-slope form.

$$y - 7 = \frac{-9}{7}(x + 3)$$

3. Rewrite the equation of the line in standard form.

$$y - 7 = \frac{-9}{7}x - \frac{27}{7}$$

$$y = \frac{-9}{7}x + \frac{22}{7}$$

$$7\left(\frac{9}{7}x + y = \frac{22}{7}\right)$$

$$9x + 7y = 22$$

Lee uses reclaimed rainwater in his garden. At 2 p.m., the water level in his rain barrel was 2.4 inches. During an afternoon rain shower, from 2 p.m. to 5 p.m., the rain fell at a steady rate. At 5 p.m., the water level in his rain barrel was 3.95. Use this information for questions 4 through 7.

4. Determine two ordered pairs on the graph of this relationship.

$(2, 2.4)$ $(5, 3.95)$

5. Use the two ordered pairs to find the slope of the line.

$$\frac{3.95 - 2.4}{5 - 2} = \frac{1.55}{3} = \frac{31}{60}$$

6. Use an ordered pair and the slope to write the equation of the line in point-slope form.

$$y - 2.4 = \frac{31}{60}(x - 2)$$

7. Use the two ordered pairs to write the equation the equation of the line in slope-intercept form.

$$y - 2.4 = \frac{31}{60}x - \frac{62}{60}$$

$$+2.4 \qquad +2.4$$

$$y = \frac{31}{60}x + \frac{41}{30}$$

A line has an x -intercept of -5 and a y -intercept of 3 . Use this information for questions 8 through 10.

8. Find the slope of the line.

$$(x_1, y_1): (-5, 0)$$

$$(x_2, y_2): (0, 3)$$

$$\frac{3 - 0}{0 - (-5)} = \frac{3}{5}$$

9. Write the equation of the line in slope-intercept form.

$$y - 0 = \frac{3}{5}(x + 5)$$

$$y = \frac{3}{5}x + 3$$

10. Write the equation of the line in standard form.

$$-\frac{3}{5}x + y = 3$$

$$5\left(\frac{3}{5}x - y = -3\right)$$

$$3x - 5y = -15$$